Communication and collaboration are vital for the data preparation process. This step involves identifying and correcting errors, filling in missing values, and resolving inconsistencies in the data. The actions you take should be discussed as a team and documented. This is especially important when working collaboratively across data teams because data engineers and data scientists often have different perspectives on how data should be prepared. For example, we have seen situations where an engineer imputed all the missing values in a column with a zero. The rationalization made sense; many zeros were already in the column, making the KPIs' values come out correctly. However, from a modeling perspective, missing data differs from zeros, especially if there are already zeros in the dataset. The approach of replacing nulls with zeros is not necessarily incorrect; it simply needs to be discussed with the downstream consumers of the data. One helpful communication tool is the column tagging functionality from the Databricks Catalog UI. See *Figure 2.5* for an example:

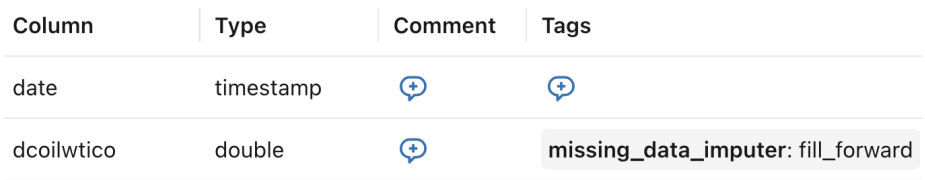


Figure 2.5 – Example of how to use tagging in a catalog to communicate the transformation performed on a column to all table users

This implementation of an incorrect imputation method also serves as an example of wanting to go back and reprocess history. Luckily, using the Medallion architecture methodology is a saving grace. In the situation mentioned previously, the chosen imputation would only be present in the Silver and Gold data layers. Meanwhile, the Bronze layer still contains the original raw data, so the source of truth is not lost, and reprocessing is possible.

One of the ways that the Databricks Platform boosts productivity and collaboration is the feature of real-time collaboration support for notebooks. This feature allows two or more people to simultaneously see and edit a notebook. The ability to pair-program virtually during the pandemic was a lifesaver for many. We're big fans of people who have worked remotely for much of our careers. Collaborative editing of a notebook is much easier than sharing code via video call. While there are many options for reviewing code, historically, reviewing code in notebooks has been difficult, particularly when committing notebooks to source control.

After completing transformations, documentation is easy using markdown in notebooks. Even if the resulting notebook is not not itself designated for production ETL, documenting the how and why of your data transformations is important for all downstream users. To read more about Databricks notebooks, see the documentation in *Further reading*.